Research Question: What are the main predictors of individual life expectancy?

Motivation

Why it matters:

- Understanding factors that influence life expectancy helps shape effective public health and social policies.
- Identifying strong predictors allows targeted interventions to improve health outcomes.
- Reduces health inequalities between countries.

Key Predictors Explored:

- Socioeconomic conditions (e.g., income, education)
- Access to healthcare
- Lifestyle behaviors (e.g., diet, immunization rates)

Significance:

- Connects statistical analysis with real-world policy impact.
- Helps policymakers prioritize strategies that extend healthy life expectancy.
- Supports global well-being initiatives.

Data Collection

Dataset: WHO Life Expectancy Data

Source:

- World Health Organization (WHO) – Global Health Observatory (GHO)
- Publicly available on Kaggle & GitHub (JayVal34, n.d.; WHO, 2015)

Sample Size & Coverage:

- ~193 countries
- 2000–2015 (combined yearly files)
- Final dataset: ~2,900 observations, 22 variables

Reliability & Accuracy:

- Official national statistics from WHO; generally reliable
- Standardized measures allow cross-country comparison

Data Appropriateness:

- Contains key macro-level variables relevant to life expectancy
- Enables analysis of predictors at the country level
- Useful for exploring socioeconomic and health-system influences on population longevity

Method of Collection:

- Health data reported annually by official country reports and national health ministries
- Indicators include life expectancy, vaccination rates, mortality, GDP, healthcare access, and social factors

Variables Include:

- Life expectancy & mortality
- Vaccination coverage (e.g., Hepatitis B)
- Economic indicators (GDP, population)
- Social and health system factors

Data Processing:

- Combined multiple yearly files into one dataset
- Removed incomplete records (e.g., missing GDP, population, Hepatitis B coverage)
- Country-level averages, not individual-level data

Methods of Analysis

Fit a multiple linear regression to predict Life Expectancy

Checked assumptions (Part 1):

- Identified skewed GDP values
- Found non-significant predictor: Year

Remedies applied:

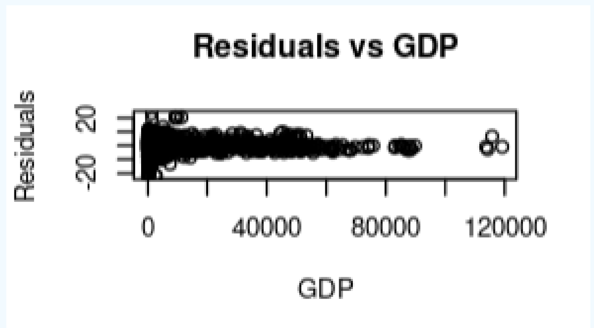
- Removed Year (model selection)
- Applied log(GDP) transformation to reduce skewness
- Influential outliers detected using Cook's Distance

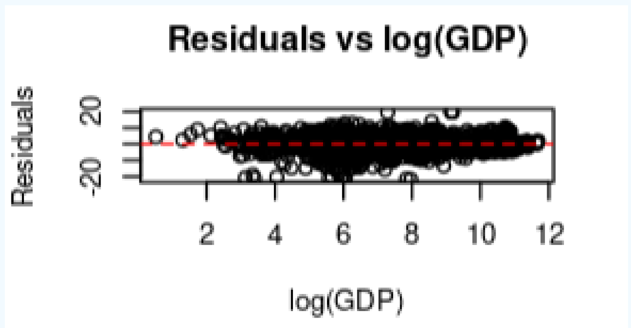
Fit improved Model:

- AIC/BIC ↓ → Better fit
- Adjusted $R^2 \approx 0.78$
- Generalized VIF (interaction model)

Assumptions rechecked for final model

```
  Status                    Adult.Mortality                          Alcohol
3.310057                           1.177219                         1.355719
     BMI                          Schooling                          log_GDP
2.695007                           1.933462                         1.350884
   Polio Income.composition.of.resources                          Status:BMI
1.111903                           1.700673                         3.409561
```

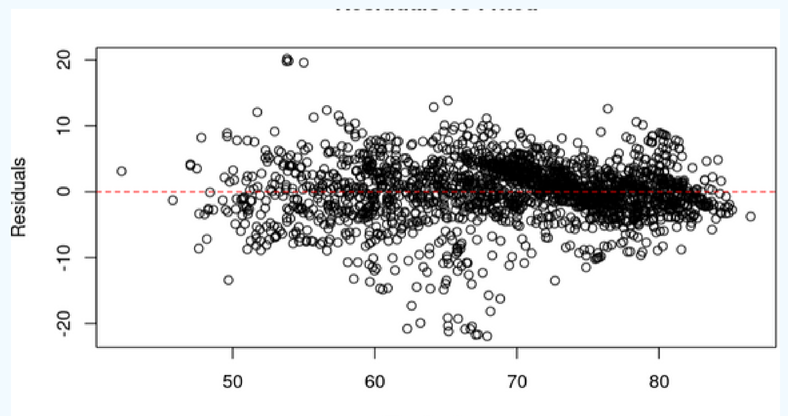


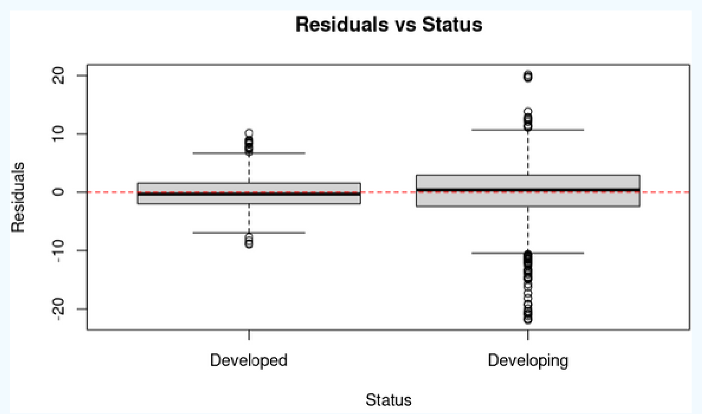


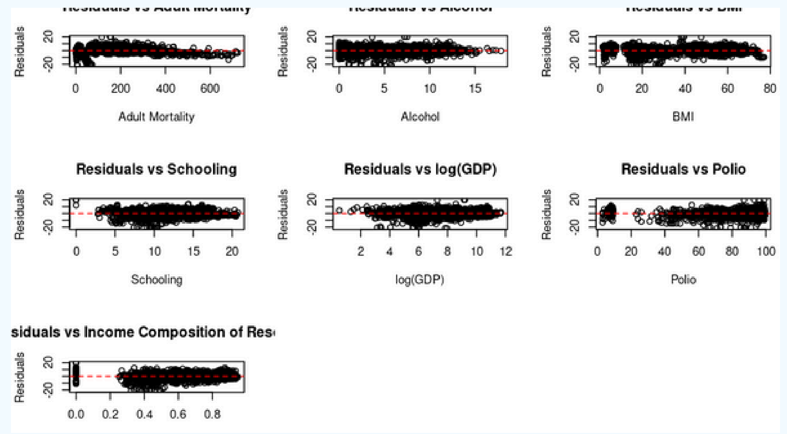
Results and Conclusions

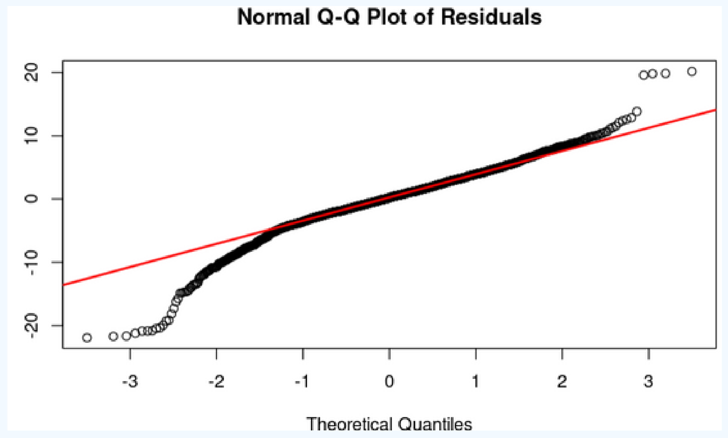
```
Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                    55.1556458  1.0955148  50.347  < 2e-16 ***
StatusDeveloping               -5.9913877  0.8295647  -7.222 7.08e-13 ***
Adult.Mortality                -0.0266132  0.0008944 -29.756  < 2e-16 ***
Alcohol                        -0.1879208  0.0327399  -5.740 1.08e-08 ***
BMI                            -0.0130096  0.0132229  -0.984    0.325
Schooling                       0.8797124  0.0579754  15.174  < 2e-16 ***
log_GDP                         0.4062277  0.0688773   5.898 4.27e-09 ***
Polio                           0.0409081  0.0047596   8.595  < 2e-16 ***
Income.composition.of.resources 8.5011690  0.7896325  10.766  < 2e-16 ***
StatusDeveloping:BMI            0.0780430  0.0148943   5.240 1.77e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.508 on 2136 degrees of freedom
  (609 observations deleted due to missingness)
Multiple R-squared:  0.7828,    Adjusted R-squared:  0.7818
F-statistic: 855.1 on 9 and 2136 DF,  p-value: < 2.2e-16

[1] 13560.13   AIC model2
[1] 12565.24   AIC final model
[1] 13623.3    BIC model2
[1] 12627.63   BIC final model
```

- **Final multiple linear regression model explains life expectancy well**
- **→ Adjusted $R^2 \approx 0.78$**
- **All predictors are statistically significant except BMI**
- **→ BMI kept because Status×BMI interaction is significant**
- **Higher adult mortality reduces life expectancy**
- **Higher schooling, log(GDP), income composition, and polio immunization increase life expectancy**
- **Developed countries show significantly higher life expectancy on average**
- **AIC/BIC decrease for final model → improved fit**
- **Residuals vs Fitted plot → linearity and constant variance satisfied**
- **Residuals vs Status: group differences captured appropriately**
- **Normal Q-Q Plot: slight tail deviation**
- **Influential points checked**









Countries with lower adult mortality, higher education levels, stronger income equality, and greater polio immunization coverage tend to experience longer lives. After GDP is adjusted using a log transformation, higher wealth is clearly associated with better health outcomes. Additionally, developed countries show substantially longer life expectancy on average.

Limitations

- **Omitted Variable Bias**: Our manual pre-selection of 9 predictors from the original 22 variables, while theoretically motivated, may have excluded other important factors. If these omitted variables are correlated with our model's predictors, it could lead to biased coefficient estimates.
- **Sample Generalizability**: The use of listwise deletion for handling missing data may limit the applicability of our findings, as the analyzed sample might not fully represent countries with less robust data reporting systems.
- **Model Specification**: Minor curvature in residual plots suggests that the linear model form may not fully capture all the underlying non-linear relationships in the data.
- **Human Bias in Variable Selection:** We prioritized theory-driven manual selection to ensure model interpretability, acknowledging this approach may introduce subjective judgment.

References

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4. JayVal34. (n.d.). *WHO Life Expectancy Data Set* [Repository]. GitHub. https://github.com/JayVal34/WHO-Life-Expectancy-Data-Set
5. World Health Organization. (2015). Life Expectancy (WHO) dataset (Global Health Observatory). https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who